

2(a)

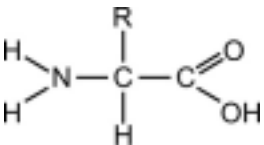
one mark each row to max 3

or = requires either root apical meristem or root cap cells statement

3

root apical meristem cells		root cap cells
(some) cells in, mitosis / different stages of cell cycle ;	or	no cells in mitosis / all cells in interphase ; A no chromosomes visible / only chromatin visible
(some) cells in, prophase / metaphase / telophase ; A chromosomes visible (in some cells)		N/A
(generally) smaller (size) ; A less cytoplasm	or	(generally) larger (size) ; A more cytoplasm
larger nucleus in proportion to, size of cell / quantity of cytoplasm ;	or	smaller nucleus in proportion to, size of cell / quantity of cytoplasm ;
nucleoli / nucleolus, (more) visible / prominent / AW ;	or	nucleoli / nucleolus, not visible / not prominent / AW ;
(generally all) cuboidal / same shape ; A rectangular / regular (shape)	or	variety of shapes / shape described ; A irregular / circular
<i>ref.</i> arrangement of cells ; e.g. rows / chains / tightly packed / compact / uniform / regular	or	<i>ref.</i> arrangement of cells ; e.g. irregular / loosely packed / more intercellular spaces
AVP ; e.g. all / more, cells have nuclei cells not, sloughing off / detaching	or	AVP ; e.g. some have no nucleus some cells, sloughing off / detaching

Question	Answer	Marks
2(b)	any three from: <i>max 1 mark (from mp1 / mp2) if response completely based on water movement within the plant</i> <i>mps 1-4: ref. to water <u>molecules</u>, needed once only / apply ecf</i> water <u>molecules</u> form hydrogen bonds ; A H bonds / H bonding cohesion between water <u>molecules</u> (in spaces) ; A cohesive force between water molecules cohesion to water <u>molecules</u>, adhering / AW, to soil, particles / clumps ; adhesion of water <u>molecules</u> to, soil, surface / particles / clumps / mucilage / polysaccharides ; A adhesive force of water <u>molecules</u> AVP ; e.g. clumps provide, areas / (small) spaces / place, in which water collects <i>idea of</i> stable environment e.g. reduced movement of water, in / around clumps I keeps water around the, soil clumps / the root tip	3
2(c)	extracellular (enzymes) ;	1
2(d)(i)	more than two / three or more / many / large number of / polymer of, monosaccharides / sugar monomers / sugar units / simple sugars / carbohydrate subunits / AW ; A macromolecule <i>for large number of</i> R if includes incorrect example of a polysaccharide, e.g. sucrose <i>plus one from:</i> joined by / AW, glycosidic bonds ; <i>idea of</i> repeating, subunits / units / monomers / AW ;	2

Question	Answer	Marks
2(d)(ii)	<i>all correct (NH₂ or COOH groups on left or right) ;</i> A H₂N  A COOH R-group / side chain, is variable or different amino acids have different, R-groups / side chains ; AW <i>this can be annotated on the diagram</i>	2
2(d)(iii)	any three from: I ref. to concentration of water / solute concentration A ψ for water potential mineral ions / nutrients, are solutes / dissolve in water ; (solutes) decrease / lower, water potential, within cell / in cytosol / in cytoplasm ; A water potential becomes, more negative / low increases / steep(er) / presence of / AW, water potential gradient ; A the difference in ψ between soil solution and cell increases A water moves from high to low water potential water follows osmotically / water enters by osmosis ; <i>in context of, presence / uptake, of mineral ions</i>	3